

Drone Arm Hand Calculations

Beam Arm Design

This document presents the hand-calculated maximum payload, maximum displacement, maximum stress, factor of safety, and material cost for the **beam arm design**, evaluated using Aluminum Alloy.

1 Geometric and Material Parameters

- Length between centers (L): 183 mm = 0.183 m
- Cross-sectional width (b): 20 mm = 0.020 m
- Cross-sectional thickness (h): 6 mm = 0.006 m
- Arm volume (CAD, net of lattice cutouts): $16,707.72 \text{ mm}^3 = 1.670772 \times 10^{-5} \text{ m}^3$
- Material: Aluminum Alloy
 - Density (ρ): 2700 kg/m³
 - Young's Modulus (E): 69 GPa = $6.9 \times 10^{10} \text{ Pa}$
 - Yield Strength (σ_{yield}): 275 MPa
 - Cost per meter: \$7.50 / m
- Motor thrust (per motor): 1.0 kg (treated as force-equivalent, per project convention)
- Motor + propeller mass: 0.075 kg
- Gravitational acceleration (g): 9.81 m/s²
- Number of motors: 4 Number of arms: 4
- Required thrust-to-weight ratio (TWR): 2.0
- Non-arm component mass: 1.0 kg

2 Maximum Payload Calculation

2.1 Arm Mass

$$m_{arm} = \rho \cdot V_{arm} = (2700 \text{ kg/m}^3)(1.670772 \times 10^{-5} \text{ m}^3) = 0.045111 \text{ kg} \quad (1)$$

$$m_{arms,total} = 4 \cdot m_{arm} = 4(0.045111 \text{ kg}) = 0.180443 \text{ kg} \quad (2)$$

2.2 Total Thrust and Maximum Allowable Weight

$$T_{total} = n_{motors} \cdot T_{permotor} = (4)(1.0 \text{ kg}) = 4 \text{ kg} \quad (3)$$

$$W_{max} = \frac{T_{total}}{TWR_{required}} = \frac{4 \text{ kg}}{2.0} = 2 \text{ kg} \quad (4)$$

2.3 Maximum Payload

$$W_{payload} = W_{max} - W_{body} - m_{arms,total} \quad (5)$$

$$W_{payload} = 2 \text{ kg} - 1.0 \text{ kg} - 0.180443 \text{ kg} = 0.8196 \text{ kg} \quad (6)$$

3 Maximum Displacement

Modeling the arm as a cantilever beam with a net tip load (thrust minus motor weight):

$$F = m_{thrust} \cdot g - m_{motor} \cdot g = (1.0)(9.81) - (0.075)(9.81) = 9.074 \text{ N} \quad (7)$$

$$I = \frac{bh^3}{12} = \frac{(0.020 \text{ m})(0.006 \text{ m})^3}{12} = 3.6 \times 10^{-10} \text{ m}^4 \quad (8)$$

$$\delta_{max} = \frac{FL^3}{3EI} = \frac{(9.074 \text{ N})(0.183 \text{ m})^3}{3(6.9 \times 10^{10} \text{ Pa})(3.6 \times 10^{-10} \text{ m}^4)} \quad (9)$$

$$\delta_{max} \approx 7.46 \times 10^{-4} \text{ m} \approx 0.746 \text{ mm} \quad (10)$$

4 Maximum Stress

$$M = F \cdot L = (9.074 \text{ N})(0.183 \text{ m}) = 1.6605 \text{ N} \cdot \text{m} \quad (11)$$

$$c = \frac{h}{2} = 0.003 \text{ m} \quad (12)$$

$$\sigma_{max} = \frac{Mc}{I} = \frac{(1.6605 \text{ N} \cdot \text{m})(0.003 \text{ m})}{3.6 \times 10^{-10} \text{ m}^4} \approx 1.384 \times 10^7 \text{ Pa} \approx 13.84 \text{ MPa} \quad (13)$$

5 Factor of Safety

$$FOS = \frac{\sigma_{yield}}{\sigma_{max}} = \frac{275 \text{ MPa}}{13.84 \text{ MPa}} \approx 19.87 \quad (14)$$

6 Material Cost

$$C_{arm} = C_{per \text{ m}} \cdot L = (\$7.50/\text{m})(0.183 \text{ m}) = \$1.3725 \quad (15)$$

$$C_{total} = 4 \cdot C_{arm} = 4(\$1.3725) = \$5.49 \quad (16)$$

7 Summary of Results

Table 1: Hand-Calculated Results vs. MATLAB FEA – Beam Arm Design, Aluminum Alloy

Quantity	Hand Calculation	MATLAB FEA
Maximum Payload	0.8196 kg	0.8195 kg
Maximum Displacement	0.746 mm	0.9788 mm
Maximum Stress	13.84 MPa	55.1848 MPa
Factor of Safety	19.87	4.89
Material Cost (4 arms)	\$5.49	–